

**ADMINISTRATIVE INFORMATION****FMI No: 25346****FMI Title: LightSpeed Pro 16 (80kW) M4 Upgrade**

| <b>FMI TYPE</b>                                      | <b>CHARGE CODES</b> | <b>IMPLEMENTATION DATES</b> |
|------------------------------------------------------|---------------------|-----------------------------|
| <input type="checkbox"/> <b>Mandatory Safety</b>     | <b>GEMS - AM:</b>   | <b>Issue Date:</b>          |
| <input checked="" type="checkbox"/> <b>Mandatory</b> | <b>GEMS - E:</b>    |                             |
| <input type="checkbox"/> <b>Mandatory on Request</b> | <b>GEMS - A:</b>    | <b>Close Date:</b>          |

|                                                            |                                                                                                                                                                                                       |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> <b>Staged FMI</b>      | <b>Customer List</b> provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. |
| <input type="checkbox"/> <b>Parts</b>                      | and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.                                                             |
| <input checked="" type="checkbox"/> <b>Parts Ordering:</b> | GEMS - Americas, Contact your Region logistics coordinator<br>GEMS - Europe, Order FMI kits from GPO/EPL Buc<br>GEMS - Asia, Contact your region logistics coordinator                                |
| <input type="checkbox"/> <b>No Parts</b>                   | are required to complete this FMI.                                                                                                                                                                    |
| <input type="checkbox"/> <b>Batch with</b>                 | _____                                                                                                                                                                                                 |
|                                                            | To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).                                     |
| <input type="checkbox"/> <b>Non-Staged</b>                 | <b>No Customer List is provided.</b> Each service location must identify affected units, and place part orders as indicated in the Special Instructions.                                              |

| <b>Technical Support</b>                                                                                                             |                                                                                                                                                     |                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>GEMS - Americas</b>                                                                                                               | <b>GEMS - Europe</b>                                                                                                                                | <b>GEMS - Asia</b>                                                                       |
| <b>GE Medical Systems</b><br><b>Insite Center</b><br><b>(414) 524-5300</b><br><b>800-321-7973</b><br><b>Milwaukee, WI 53201, USA</b> | <b>GEMS - E / ESC</b><br><b>BP34</b><br><b>F 78533 BUC CEDEX, FRANCE</b><br><b>Tel: (33 1) 39 20 00 07 or ESC</b><br><b>Fax: (33 1) 30 70 99 70</b> | <b>Asia Support Center</b><br><b>Phone: 81-426-56-0033</b><br><b>Fax: 81-426-48-2905</b> |

**Special Instructions:****Unit Tracked: 2357739-3****FMI Instructions Part No.: 5115297****Do NOT leave unsecured on site**

# **FMI** Completion Certification Report

## Mail To

**GEMS - Americas**

**GEMS - Europe**

G.E. Medical Systems  
Attn: FMI Admin. W-523  
P.O. Box 414  
Milwaukee, WI 53201

GEMS - Europe  
FMI Admin. 322  
BP34  
F78533 Buc Cedex, France

Send to your  
Country FMI or Service  
Administrator

FMI No.: 25346

FMI Title: LightSpeed Pro 16 (80kW) M4 Upgrade

Customer Full Name: \_\_\_\_\_

Number and Street Address: \_\_\_\_\_

City and Country: \_\_\_\_\_

Country Code

Site Number

System I.D.

### Unit Tracked:

| Model Number | Serial Number | Compl. Date<br>DD/MM/YY | FMI Comp.<br>Code | Service Hours |        |
|--------------|---------------|-------------------------|-------------------|---------------|--------|
|              |               |                         |                   | Labor         | Travel |
|              |               |                         |                   |               |        |
|              |               |                         |                   |               |        |
|              |               |                         |                   |               |        |
|              |               |                         |                   |               |        |

### FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

### Customer Refusal or Equipment Altered - Modification Pending.

Installer  
Comments \_\_\_\_\_

## Effectivity

All LightSpeed Pro (80kW) systems produce to date (May 10, 2004).

## Purpose

The purpose of this document is to upgrade all LightSpeed Pro16 80kW sites to production software and hardware. Additionally, there will be a change made to the TGP circuit board and Performix Pro tube (addition of the Smart ID circuit board to the tube structure).

The Smart ID circuit board kit is provided in this FMI Kit for the Performix Pro tube. If you do not have a Smart ID kit, or the circuit board is not functioning properly please perform the rest of this FMI and contact:

Jim Dodge, Mechanical Engineering Team Lead, FCT Source– cell # (262) 424-2848, e-mail: James.Dodge@med.ge.com

Jim will provide this part to you ASAP.

## Furnished Materials

| Qty | Description                             | --Kit -- | Part #                 |
|-----|-----------------------------------------|----------|------------------------|
| 1   | GRE LFC Software Installation Procedure |          | 2378261-100<br>Rev.10  |
| 1   | FMI Document                            |          | 5115297-100            |
| 1   | LightSpeed 5.X Tips and Workarounds     |          | 2407441-100            |
| 1   | CT LightSpeed Series CD                 |          | 2369740-247            |
| 1   | LightSpeed Pro & RT Software            |          | 5114319                |
| 1   | Smart ID Kit                            |          | 5115466                |
| 2   | PROMs for TGP board                     |          | 2371062-4<br>2362522-5 |
| 1   | Dynamic Brake Fuse                      |          | 46-170021P106          |

## Tools Required

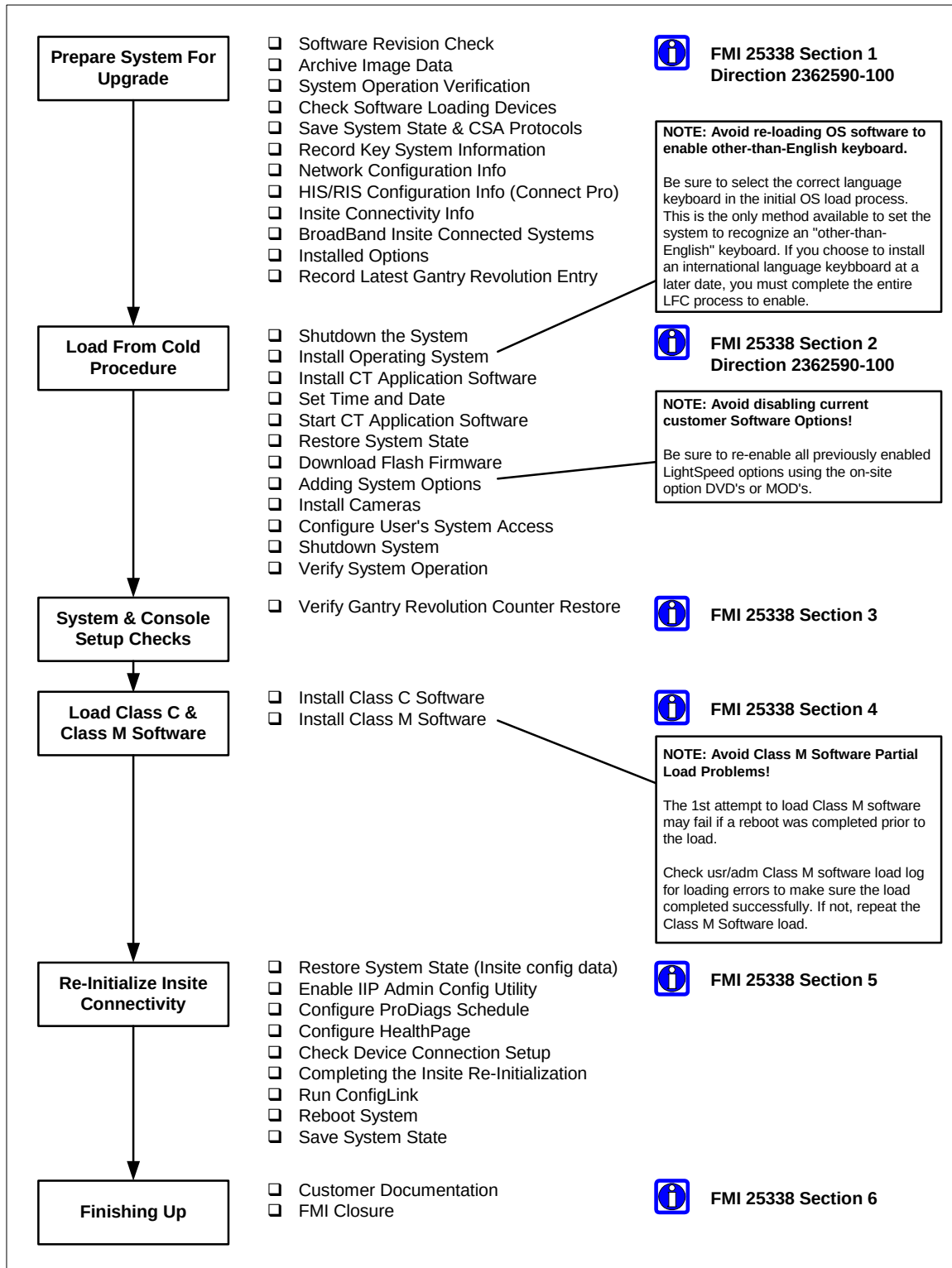
| Qty | Description                     | Part #  |
|-----|---------------------------------|---------|
| 1   | LightSpeed 5.X Class C Software | 5114786 |
| 1   | LightSpeed 5.X Class M Software | 5114787 |

## ***People Power***

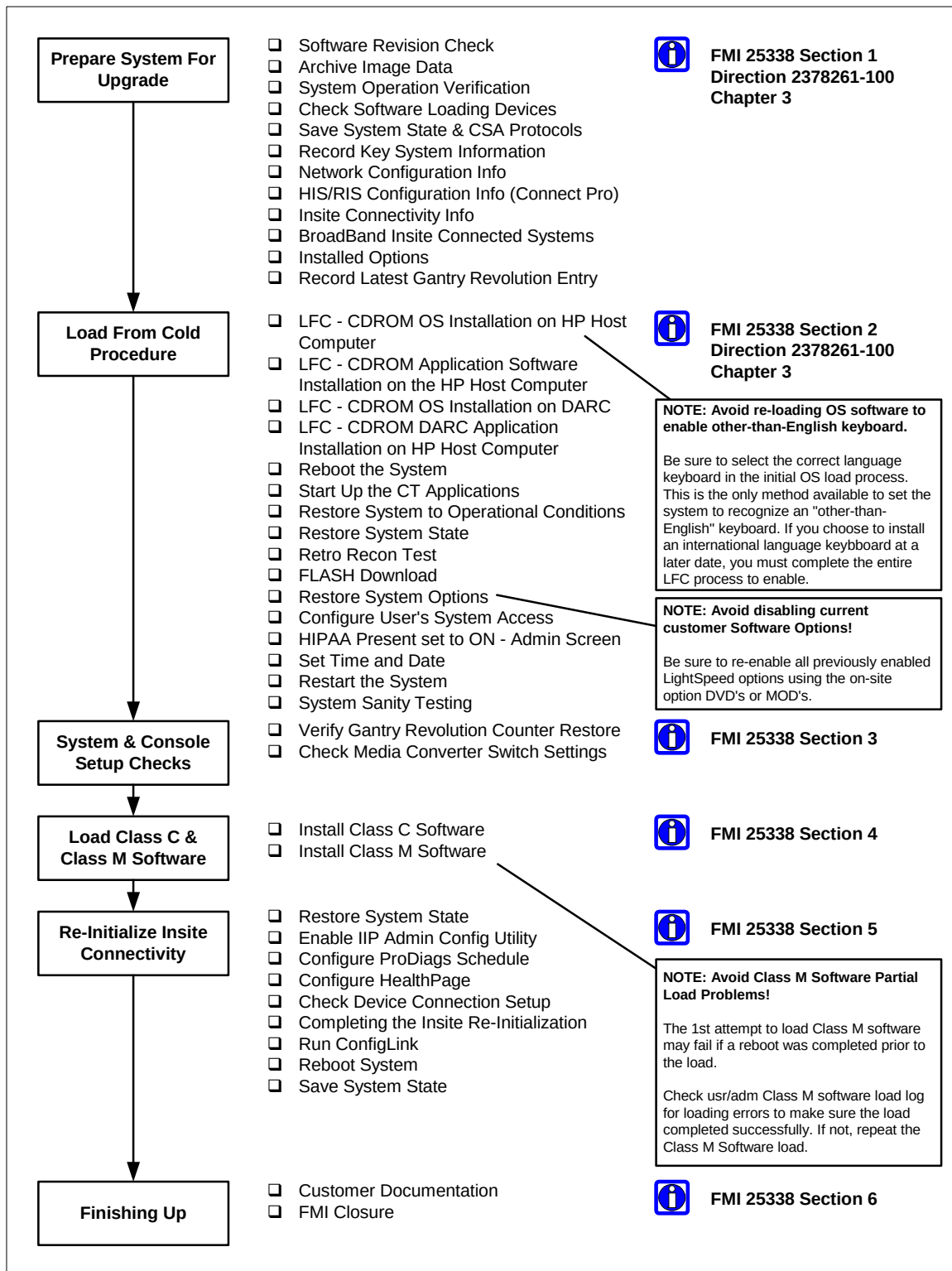
Travel: 1 Hour

Labor: 4 Hours

## Steering Guide for Linux Pegasus (GOC2)



## Steering Guide for GRE (GOC3)



## Procedure

This FMI includes the following procedures:

- Smart ID Circuit Board Installation.
- Axial Dynamic Brake Fuse Replacement
- TGP PROM change
- Preparing the System for Software Upgrade
- Load From Cold

### 1 Smart ID Circuit Board Installation (LightSpeed Pro 16 80kW – Performix Pro Tube)

The following instructions detail the field process to provide SmartID functionality for Performix Pro (D3634T) X-Ray tubes built without SmartID functionality.

#### Preliminary Notes:

Refer to drawing below as needed.

#### Instruction:

- 1) **Remove power to the gantry using lockout / tagout.**
- 2) Remove right side gantry cover
- 3) Rotate gantry such that the x-ray tube is at the 3 o'clock position

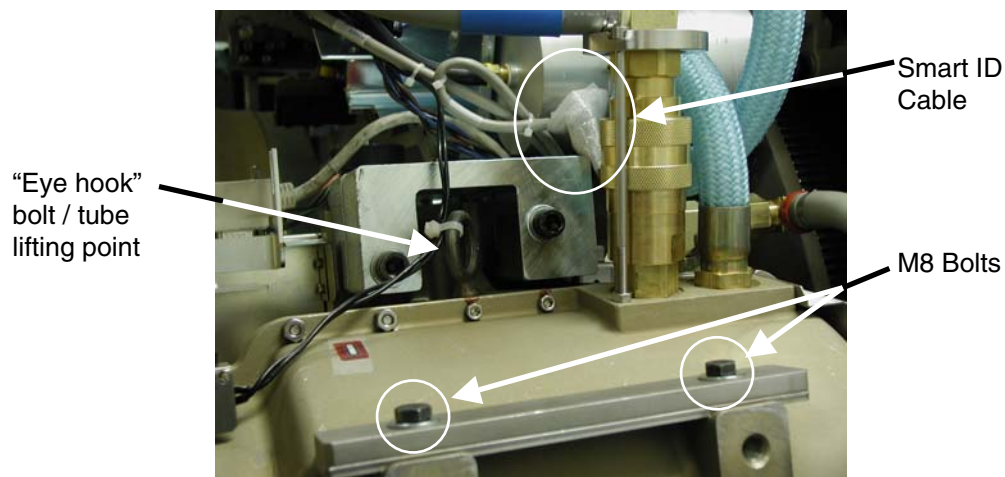


Fig 1 – Performix Pro Config without Smart ID

- 4) Remove the two tube balance weight M8 mounting bolts and washers on the topside of the tube casing. Keep hardware – you will reuse it.
- 5) Place SmartID assembly 5115466 over the tube weights and reinstall the M8 bolts/washers using a drop of blue loctite (242) on the bolt threads.
- 6) Torque the two M8 bolts to 20N-m [14.8 lb-ft / 177 lb-in / 204 kg-cm].
- 7) Connect Smart ID cable to Smart ID board. You will find this cable coming from the Jedi Inverter, connector J5, which also connects to the heat exchanger power i/f board. The Smart ID cable is the disconnected portion of the Y cable. See figure 1 if necessary.
- 8) Reattach tie-wrap for tube thermal wires to tube lifting hook and reattach any other tie-wraps you may have cut during the procedure.

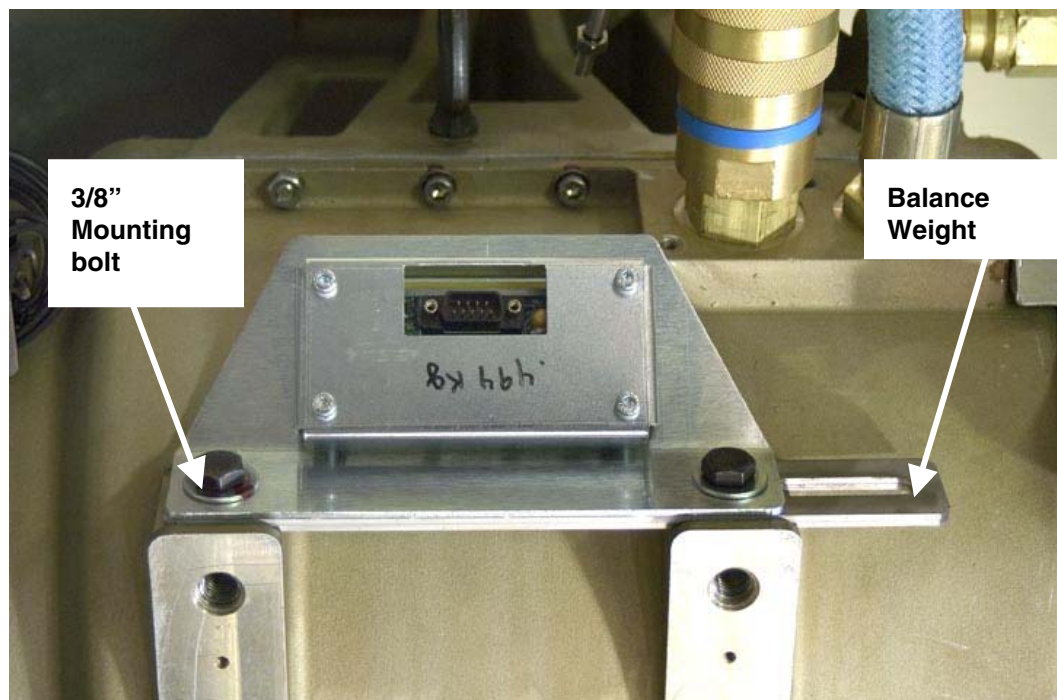


Fig 2 – Mock drawing of Smart ID installation points on Performix Pro tube

## 2 TGP PROM Change

Two PROM devices on the TGP board need to be changed to update the code on this board. The biggest issue this solves is a “trigger jitter” issue, which was present in older versions of these PROM devices. The procedure below details how to change out these devices.

**Warning: Please be careful when removing and installing these devices as they can be damaged very easily.**

- 1) Verify that the **power to the gantry is still removed.**
- 2) Wear a Wrist Strap and follow proper Anti-Static procedures
- 3) Remove the Sheet Metal Cover on the TGP and MSUB enclosure
- 4) Using a small flathead screw driver or a pair of tweezers, gently pry up I.C. Chips U11 and U18 on the TGP Board ( Upper board in this chassis ). Pry on the Notched Side ( pin 1 end ) and then the opposite end switching back and forth a few times to avoid damage to the IC Socket. These Chips are 8 pin DIP Packages and should have small white stickers on them

**Note:** Be careful and patient while removing these chips, as damage to the circuit board mounted IC socket results in the need to replace the entire circuit board.

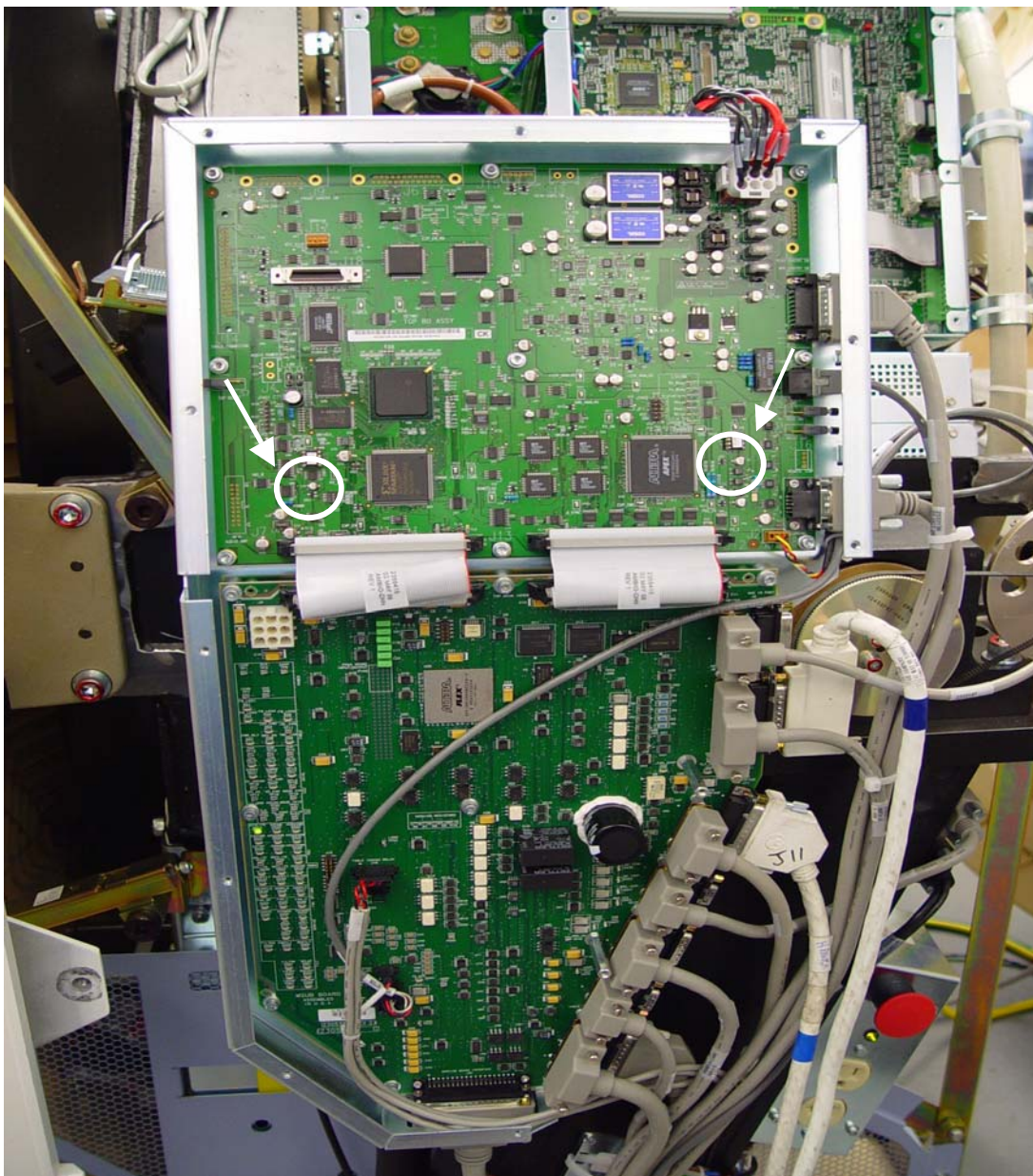
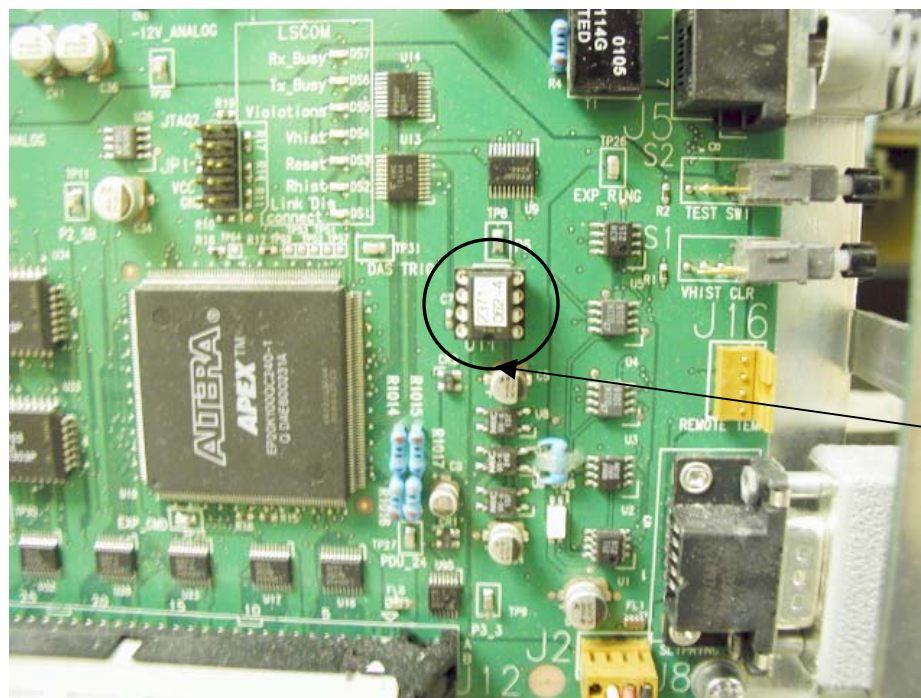


Fig 3: TGP Board on Gantry View

5) Install the replacement chip for U11 in the TGP making sure that the notch on the chip aligns with the notch in the Socket (should be positioned downwards towards the floor). The replacement chip for U11 has a sticker on it labeled 2371062-4. This is the Serial PROM for the Altera Chip on the TGP included in the FMI.



Notch Is  
pointed  
down-ward

Fig 4: Chip location / installation

6) Install the replacement chip for U18 in the TGP making sure that the notch on the chip aligns with the notch in the socket (should be positioned downwards towards the floor). The replacement chip for U18 has a sticker on it labeled 2362522-5. This is the Serial PROM for the Xilinx Chip on the TGP.

7) Using the Sticker / Label provided change the version of the TGP. The kit provides a small sticker that has the following text on it **2290138-5** use this sticker to update the Bar Code Label of the TGP Board. Place the new sticker so it covers up the old board number listed here **2290138-04**. Position the new sticker so it covers only the old text and leave the Bar Code pattern uncovered.

### 3 Replace Dynamic Brake Fuse

The Axial Dynamic Brake fuse F2 exists to protect the transformer. This fuse can blow due to inrush current for 50Hz systems due to slightly higher operating currents at the lower frequency. The Littlefuse manufactured fuses have a tendency to blow sooner than the Bussman fuses of the same rating. To resolve this issue, the 7A fuse is being replaced with an 8A fuse (46-170021P106 Bussman only). The Hpower Service Fuse kit (2218570-2) is being changed to have the Bussman only 8A fuse (46-170021p106).

Procedure:

1. Remove the rear cover.

2. Replace the 7A fuse, shown in figure 5 below, with the 8A fuse supplied in the FMI kit. Using a black marker, relabel near the fuse holder to state "8A".

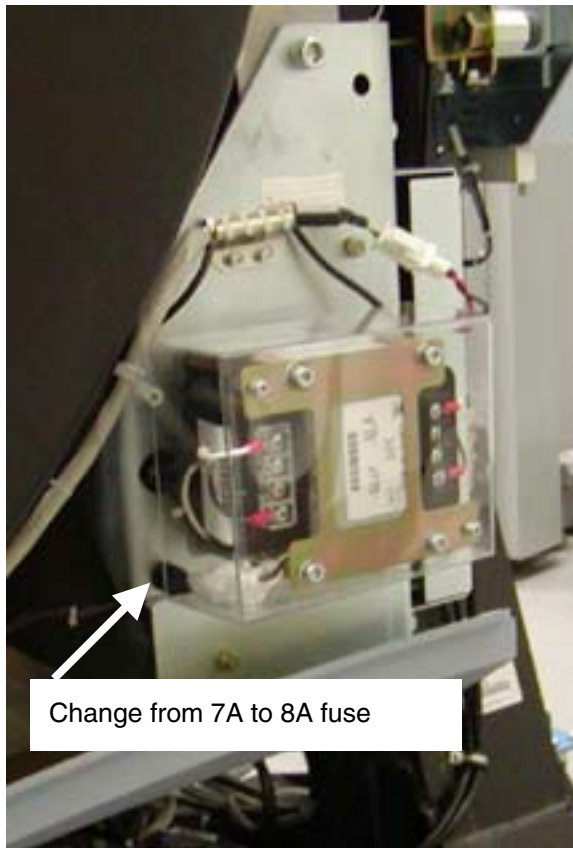


Figure 5: Dynamic Brake Assembly

3. Restore power to the system.
4. Prescribe a test scan using the service generic protocol 42.1, accept the protocol and scan. If you receive errors and can't scan due to an Axial Drive issue, check the fuse.
5. Disable Axial Drive and reassemble covers. Enable Axial drive prior to putting on the right side cover. Prepare System for Software Upgrade

### 3.1 System Operation Verification

Put all the covers back and power up the system.

Set-up and perform at least one axial scan and one helical scan to verify that the system operated properly before beginning the software upgrade

### 3.2 Software Revision Check

Check the current version of CT Application Software on the system using the showprods command.

1. Open a shell.

2. At the prompt enter the following command

```
showprods | grep Helios_OC ENTER
```

### 3.3 Archive Image Data

**The following procedures (Software Load From Cold) will destroy all image data on the system. Prior to starting the software load, make sure that the customer has reconstructed and archived all images before proceeding.**

### 3.4 Check Software Loading Devices

Since the CDROM Drives on Linux systems are seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of these devices before you begin the LFC procedure.

**Reference:** Direction 2378261-100, GRE Software Installation Appendix E

- |             |                                                  |
|-------------|--------------------------------------------------|
| Section 2.1 | Remote Shell – Check Internal Network            |
| Section 2.2 | HP Computer CDROM – Mount and Unmount Process    |
| Section 2.3 | Peripheral Tower DVD - Mount and Unmount Process |
| Section 2.4 | Peripheral Tower DVD – Initializing a DVD        |
| Section 2.5 | Peripheral Tower MOD - Mount and Unmount Process |

### 3.5 Save System State & CSA Protocols

This section details the procedure for archiving the system critical files (System State) and adding to them the customer protocol settings that previously were not included in the System State archive process. This newly created System State will be used during the upgrade Load From Cold Procedure.

**Reference:** Direction 2378261-100, GRE Software Installation

- |               |                        |
|---------------|------------------------|
| Section 3.2   | Save System State      |
| Section 3.2.2 | CSA Protocol Retention |

### 3.6 Record Key System Configuration Information

To ensure all system configuration information can be restored after the Load From Cold process, record the following parameters. Although the Save System State utility saves most of these parameters, recording this information provides a means to check that these parameters are properly restored after the Restore State Process.

### 3.7 Network Configuration Info (found in /usr/g/config INFO)

Check CT LightSpeed Network Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

```
cd /usr/g/config ENTER
```

```
more INFO ENTER
```

From the usr/g/config INFO file, record the following network configuration parameters:

| Parameter            | Line Name in /usr/g/config INFO | Record Parameter Value | Notes                                                                                   |
|----------------------|---------------------------------|------------------------|-----------------------------------------------------------------------------------------|
| Hospital Name        | HOSPITAL_NAME                   |                        | Can have apostrophe with IIP 3.2                                                        |
| Hostname             | GATEWAY_HOSTNAME                |                        |                                                                                         |
| System IP Address    | GATEWAY_IP                      | ____.____.____.____    |                                                                                         |
| Broadcast IP Address | GATEWAY_BROADCAST               | ____.____.____.____    |                                                                                         |
| Netmask IP Address   | GATEWAY_NETMASK                 | ____.____.____.____    |                                                                                         |
| Default Gateway      | GATEWAY_DEFAULT                 | ____.____.____.____    | May have been used as the Insite Default Gateway in previous BroadBand Insite Checkout. |
| NIS Server           | NIS_SERVER                      | ____.____.____.____    |                                                                                         |

### 3.8 HIS/RIS Configuration Info (found in /usr/g/config INFO)

**NOTE:** If you do not have HIS/RIS, you can skip this section.

Check CT LightSpeed NIS/RIS Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

```
cd /usr/g/config ENTER
```

```
more INFO ENTER
```

From the usr/g/config INFO file, record the following HIS/RIS configuration parameters:

| Parameter          | Line Name in /usr/g/config INFO | Record Parameter Value | Notes |
|--------------------|---------------------------------|------------------------|-------|
| HIS/RIS Title      | HRCTTITLE                       |                        |       |
| HI/RIS Port Number | HRPORTNUM                       |                        |       |
| HIS/RIS AE Title   | HRAETITLE                       |                        |       |
| HIS/RIS IP Address | HRIPADDR                        | ____.____.____.____    |       |

### 3.9 Insite Connectivity Info (see .insiteINFO file)

NOTE: If you don't have Insite, you can skip this section.

1. Open a UNIX shell.
2. At the prompt enter the following command

```
cd /usr/g/insite ENTER
```

```
more .insiteINFO ENTER
```

From the usr/g/insite .insiteINFO file, record the following Insite Connectivity configuration parameters:

| Parameter                   | Line Name in<br>/usr/g/insite .insiteINFO | Record Parameter<br>Value | Notes                       |
|-----------------------------|-------------------------------------------|---------------------------|-----------------------------|
| Connection Type             | INS_ConnectionType                        |                           |                             |
| Modem Type                  | INS_ModemType                             |                           | Modem Connectivity Only     |
| Modem Phone Number          | INS_PhoneNumber                           |                           | Modem Connectivity Only     |
| Modem Dial Prefix           | INS_DialPrefix                            |                           | Modem Connectivity Only     |
| Modem Tone or Pulse Setting | INS_ToneOrPulse                           |                           | Modem Connectivity Only     |
| Modem Country               | INS_Country                               |                           | Modem Connectivity Only     |
| Modem String                | INS_ModemString                           |                           | Modem Connectivity Only     |
| Modem TTY Port              | INS_TtyPort                               |                           | Modem Connectivity Only     |
| Modem TTY Speed             | INS_TtySpeed                              |                           | Modem Connectivity Only     |
| BroadBand Insite Gateway IP | INS_GatewayIP                             | ---.---.---.---           | BroadBand Connectivity Only |
| BroadBand Gateway Mask      | INS_GatewayMASK                           | ---.---.---.---           | BroadBand Connectivity Only |

### 3.10 BroadBand Insite Connected Systems

If your system has already been Insite Checked Out as a Broadband system, you will need to re-enter the Insite Gateway IP Address into the IIP Admin Configuration utility when reconfiguring Insite after the Load From Cold process.

To ensure that you enter the correct IP Address of the Insite dedicated Gateway, confirm this IP Address with one of the following resources:

- Ask the Hospital Network Administrator or IT contact.
- In the United States, check for the IP Address at the following web site or by calling the regular checkout number (800-321-7937):

<http://gein2.med.ge.com/vpni/secure/index.jsp>

- In Europe, contact the regular toll free number to reach the OLCE.

IP Address: \_\_\_\_\_

Perform this investigation prior to arriving on site.

### 3.11 Installed Options (found using swokinstall -p)

Check CT LightSpeed Pro 16 Options Settings as follows:

- 1) Open a UNIX shell.
- 2) At the prompt enter the following command

**swokinstall -p** **ENTER**

From the swokinstall -p output, record which of the following LightSpeed 16 system options are installed on the system:

| Line Name in output file of swokinstall -p | Enabled (Y/N) | Permanent | Flex Trial | Standard LightSpeed 16 Option or Purchased Option |
|--------------------------------------------|---------------|-----------|------------|---------------------------------------------------|
| 0.4 Speed                                  |               |           |            | Standard                                          |
| Smart Prep                                 |               |           |            | Standard                                          |
| Smart Speed                                |               |           |            | Standard                                          |
| Helical Tilt                               |               |           |            | Standard                                          |
| Direct-3D                                  |               |           |            | Standard                                          |
| VariViewer                                 |               |           |            | Standard                                          |
| AutomA                                     |               |           |            | Standard                                          |
| Power440                                   |               |           |            | Standard                                          |
| Tube6_3MHU                                 |               |           |            | Standard                                          |
| 3000 Image Series                          |               |           |            | Standard                                          |
| Patient-16-slice                           |               |           |            | Standard                                          |
| Thin Twin Helical                          |               |           |            | Standard                                          |
| Connect Pro                                |               |           |            | Purchased Option                                  |

| Line Name in output file of swokinstall -p | Enabled (Y/N) | Permanent | Flex Trial | Standard LightSpeed 16 Option or Purchased Option |
|--------------------------------------------|---------------|-----------|------------|---------------------------------------------------|
| DentaScan                                  |               |           |            | Purchased Option                                  |
| Smart Score                                |               |           |            | Purchased Option                                  |
| Smart Score Pro                            |               |           |            | Purchased Option                                  |
| CardIQ SnapShot                            |               |           |            | Purchased Option                                  |
| Smart Step                                 |               |           |            | Purchased Option                                  |
| CTPerfusion2                               |               |           |            | Purchased Option                                  |
| VolumeViewer                               |               |           |            | Purchased Option                                  |
| AdvVesselAnalysis                          |               |           |            | Purchased Option                                  |
| CTColono                                   |               |           |            | Purchased Option                                  |
| CardIQ2                                    |               |           |            | Purchased Option                                  |
|                                            |               |           |            |                                                   |

**NOTE:** Make sure that for each Option enabled, there is an on-site DVD that contains the key(s) for the Options presently enabled. Without the DVD keys, you will not be able to restore options on the system that were previously enabled.

## 4 Load From Cold Procedure

After system operation has been verified, all images have been archived, a Save System State has been generated, and a record of key system configuration parameter has been completed, you are ready to upgrade the system software.

### 4.1 Shutdown Through Start CT Applications Steps

#### *Linux/GRE Consoles*

Using the LightSpeed GRE Software Installation Procedure document provided in this evaluation kit (23786261-100 Rev 09.2 Provided in the evaluation kit) and the software included in this document perform a software load.

**Reference:** Direction 23786261-100 Rev 09.2 Provided in the evaluation kit, GRE Software Installation

|               |                                                                       |
|---------------|-----------------------------------------------------------------------|
| Section 3.3   | Load From Cold Installation Procedures                                |
| Section 3.3.1 | LFC – CDROM OS Installation on HP Host Computer                       |
| Section 3.3.2 | LFC – CDROM Application Software Installation on the HP Host Computer |
| Section 3.3.3 | LFC – CDROM OS Installation on DARC                                   |
| Section 3.3.4 | LFC – CDROM DARC Application Installation on HP Host Computer         |
| Section 3.4   | Reboot the System                                                     |

Section 3.5 Start Up the CT Applications

## 4.2 Restore System To Operational Conditions

### *Linux/GRE Consoles*

Using the LightSpeed GRE Software Installation Procedure document provided in this evaluation kit (23786261-100 Rev 09.2 Provided in the evaluation kit) and the software included in this document perform a restore of the system to operational conditions.

**Reference:** Direction 23786261-100 Rev 09.2 Provided in the evaluation kit, GRE Software Installation

Section 3.8 Restore System State (and CSA Protocols)

Section 3.9 Retro Recon Test

Section 3.10 FLASH Download

**NOTE:** Make sure that the Key System Information recorded in Section 1 (Network Configuration Info and HIS/RIS Configuration Info) is correctly set as recorded. If not, use the following to:

- reconfig utility to reset Network Configuration Info.
- Connect Pro utility to reset the HIS/RIS Configuration Info.

## 4.3 Restore System Options

Using the record of installed options prior to the software load, install all the options using the DVD keys for each, as well as install new [HP100/80 Power Option key on the system](#).

**Reference:** Direction 2378261-100, GRE Software Installation

Section 3.12 Install Options (make sure you install new option Hi Power – DVD Provided in a package)

## 4.4 Complete the Load From Cold Process

**Reference:** Direction 2378261-100, GRE Software Installation

Section 3.6 Configure Users System Access

Section 3.7 HIPAA Present set to ON – Admin Screen

Section 3.13 Set Time and Date

Section 3.14 Restart the System

Section 3.15 System Sanity Scanning

**NOTE: AVOID LOSING INSITE CONNECTIVITY CONFIGURATION DATA!!**

**DO NOT SAVE SYSTEM STATE UNTIL YOU'VE COMPLETED LOADING CLASS C AND CLASS M SOFTWARE AND RE-INITIALIZED INSITE CONNECTIVITY. PERFORMING THE SAVE STATE PROCESS PRIOR TO LOADING CLASS C AND CLASS M SOFTWARE MAY CAUSE THE LOSS OF CRITICAL INSITE CONFIGURATION PARAMETERS AND RESULT IN THE NEED TO PERFORM AN INSITE CHECKOUT TO RESTORE CONNECTIVITY.**

## 5.....Load Class C and Class M Software

### Notice: Avoid Partial Load of Class M Software

To ensure a proper load of the Class M Software, load Class C Software and immediately load Class M Software without rebooting the system between CD loads. The Class M Software CD load will fail to mount if not completed in this sequence. If the Class M Software install log exhibits Errors, usually a repeat attempt at loading Class M Software CD succeeds.

### 5.1 Load Class C Software

Load provided Class C software as follows:

1. Open a shell.
2. Enter root as superuser:
  - a. At the ctuser prompt, type: `su -` **ENTER**
  - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Insert the Class C software CD into HP computer DVD Drive. Wait for the disk read to STOP.
4. Install the Class C software on the system. At the root prompt, type:  
`install.classC` **ENTER**
5. Close Class C install reporting shell.
6. Verify Class C is properly loaded on the system.
  - a. At the root prompt, type: `cd /var/adm` **ENTER**
  - b. At the root prompt, type: `ls` **ENTER**
  - c. Find the `install.classc.log` file. If file found, this means the Class C software load completed.
  - d. At the root prompt, type:  
`more install.classc.log.<yyyymmddhhmmss>`  
Refer to Appendix A for an example of a successful Class C software load.
7. Remove the Class C software CD from HP computer DVD Drive.

### 5.2 Load Class M Software

Load provided Class M software as follows:

1. Open a shell.
2. Enter root as superuser:
  - a. At the ctuser prompt, type: `su -` **ENTER**
  - b. At the Password prompt, type: `#bigguy` **ENTER**

3. Insert the Class M software CD into the HP computer DVD drive. Wait for the disk read to STOP.
4. Install the Class M software on the system. At the root prompt, type:  
`install.classM` **ENTER**
5. Verify Class M is properly loaded on the system.
  - a. At the root prompt, type: `cd /var/adm` **ENTER**
  - b. At the root prompt, type: `ls` **ENTER**
  - c. Find the install.classM.log file. If file found, this means the Class M software load completed.
  - d. At the root prompt, type:  
`more install.classM.log.<yyyymmddhhmmss>`  
Refer to Appendix A for an example of a successful Class M software load.
6. Remove the Class M software CD from HP computer DVD Drive.

## 6 ..... Re-Initializing Insite Connectivity

This release of software contains a new IIP release (IIP 3.2-Linux). Changes in this IIP release are primarily in the following areas:

- Updates to support Multi-Tech modem firmware revision 8.19M
- Removal of IIP\_DialPrefix ProDiags task
- Replacement of current iipHealthPage ProDiags task with a new Insite Data Mining task
- Auto Route to AutoSC in Level 5 setup

In addition, the Network Configuration Tab in the Insite Interactive Administration utility has been modified to eliminate the Use Default Gateway selection.

To ensure these IIP 3.2 features are correctly set, an Insite Checkout is NOT required. Because the basic Insite configuration information on the system should be correct from the original Insite Checkout, in most cases, it is only necessary to run ConfigLink via exiting the IIP Admin Configuration utility Insite Checkout Tab Screen.

### 6.1 Restore System State To Retrieve Class M Configuration Parameters

Once Class M software has been loaded, you must repeat a Restore System State to recover all previous Insite and Class M configuration parameters.

**Reference:** Direction 2378261-100, GRE Software Installation

Section 3.8      Restore System State

**NOTE:** Make sure that the Key System Information recorded in Section 1 (Insite Connectivity Info) is correctly set as recorded. If not, use the IIP Admin Configuration utility to reset Network Configuration Info as described in the following procedures.

### 6.2 Enable Insite Interactive Platform Configuration Utility

See Figure 2, Insite Interactive Platform Configuration Utility Gaining Access and Selecting ProDiags Utility.

Enable the Insite Interactive Platform Configuration utility as follows:

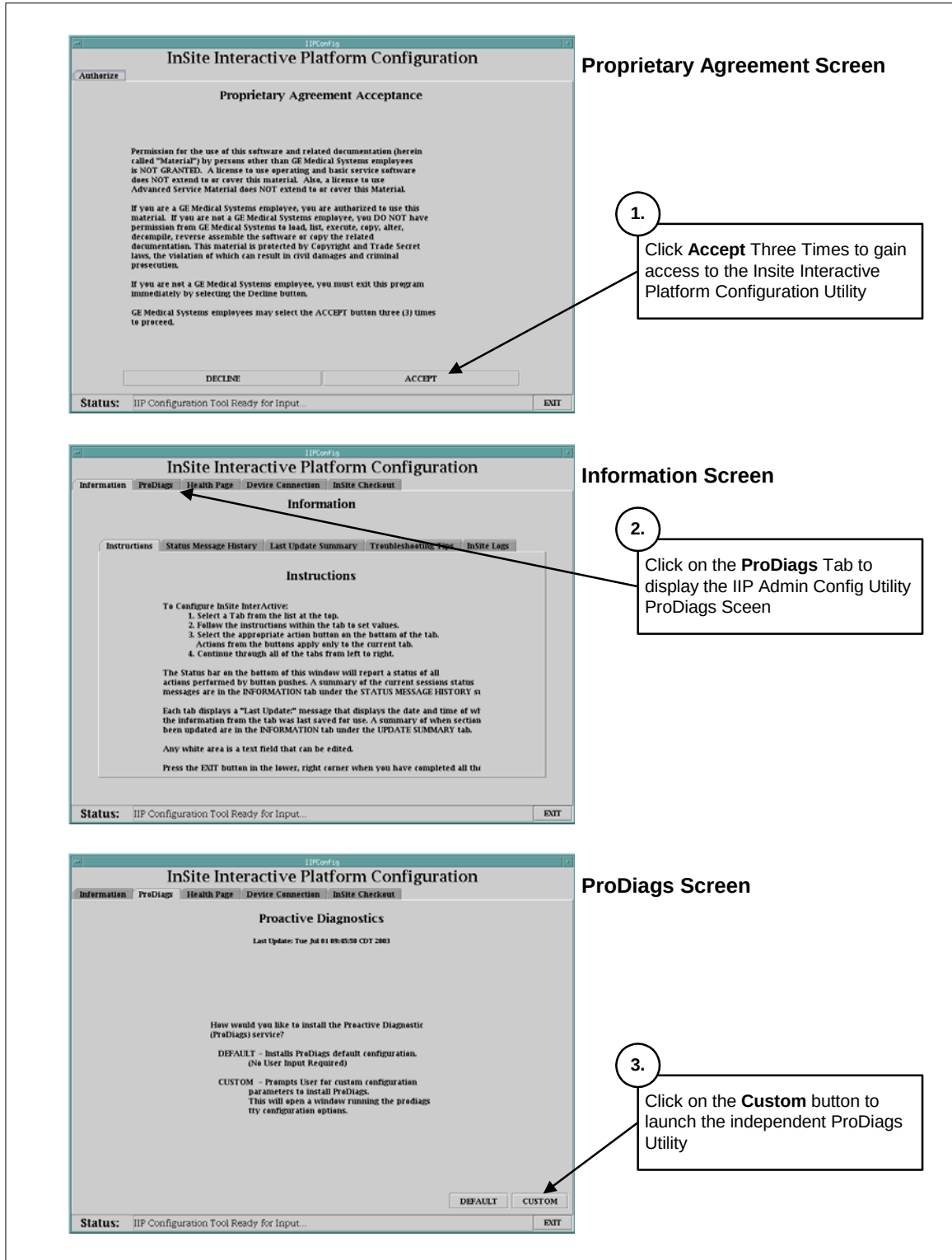
1. Open a shell.
2. Enter root as superuser:
  - a. At the ctuser prompt, type: `su -` **ENTER**
  - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Launch the IIP Admin Configuration utility. At the root prompt, type:

`iipadmin config` **ENTER**

The system displays the Insite Interactive Platform Configuration utility with the Proprietary Agreement Acceptance Screen.

4. Click on **ACCEPT** three times to display the Information Screen.
5. Click on the **ProDiags** Tab to display the ProDiags Screen.
6. Click on the **Custom** button to launch the independent ProDiags utility as shown in Figure 3.

Figure 2  
Insite Interactive Platform Configuration Utility  
Gaining Access and Selecting ProDiags Utility



### 6.3 Configure ProDiags Schedule

With this release of software, the IIP 3.2 package contains a new data collection package that pushes gesyslog data and a basic system configuration file to the On Line Center RDA Database. This IDM package contains two new ProDiags tasks (idm and idm\_send\_data) that must be configured to match the site's operational usage.

The default ProDiags schedule provided in this release should be adequate for most systems. However, we recommend that you review this schedule and update according to your particular site's normal operation.

#### Updating ProDiags Scheduled Tasks

See Figure 3 and refer to detailed steps on the following page.

**Figure 3**

#### ProDiags Graphical Utility

#### Schedule Tab – Modifying Scheduled Tasks

**1.** Click on **Hour** and **Minute** fields of **idm\_send\_data** task and select an hour and minute from the pull down lists between 9 AM and 5 PM.

After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

**2.** Click on **Hour** and **Minute** fields of **idm** task and select an hour and minute from the pull down lists to schedule 5 minutes before the **idm\_send\_data** task.

After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

**3.** When **idm** and **idm\_send\_data** Hour and Minute selections are complete, click on **Apply** to finalize the schedule updates.

Click **Yes** on the "Are you sure you want to apply changes?" pop up.

**4.** Close the ProDiags utility browser by clicking on **Close** from the upper left browser window pull down menu.

Note: If

Use the following sequence to update the scheduled time (Hour and Minute) of the `idm` and `idm_send_data` tasks:

ProDiags `idm_send_data` task

- Click on the **`idm_send_data`** task **Hour** field.
- From the pull down list, select an hour that the system is normally on (typically between 09:00 and 17:00).
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.
- Click on the **`idm_send_data`** task **Minute** field.
- From the pull down list, select a minute (preferably an odd minute such as 3, 14, 22, etc).
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.

ProDiags `idm` task: Set up the `idm` task to run five minutes before the `idm_send_data` task.

- Click on the **`idm`** task **Hour** field.
- From the pull down list, select an hour that sets the `idm` task to run five minutes before the `idm_send_data` task.
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.
- Click on the **`idm_send_data`** task **Hour** field.
- From the pull down list, select a minute that sets the `idm` task to run five minutes before the `idm_send_data` task.
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.

When completed with the `idm` and `idm_send_data` modifications, click on the **APPLY** button at the bottom of the the Schedule Tab. On the “Apply change confirmation” pop-up, click on **YES**.

**NOTICE:** If the schedule change failed due to a permissions denied error, execute the following change-mode command to make the file named `bg_schedule_data` read/writable:

- With the ProDiags GUI still open, open another shell
- Become root user:  
At the `ctuser` prompt, type: `su -` **ENTER**  
At the Password prompt, type: `#bigguy` **ENTER**
- Set Permissions for `bg_schedule_data`:  
type: `cd /usr/g/insite/ProDiags.schedule`  
type: `chmod 777 bg_schedule_data`

Repeat the Apply click on the ProDiags Schedule Tab. The update should occur without failures.

Close the ProDiags utility browser by selecting **Close** from the upper left browser window pull down menu. The ProDiags browser should close and the Insite Interactive Platform Configuration utility should display the HealthPage Tab as shown in Figure 4.

## 6.4 Configure HealthPage

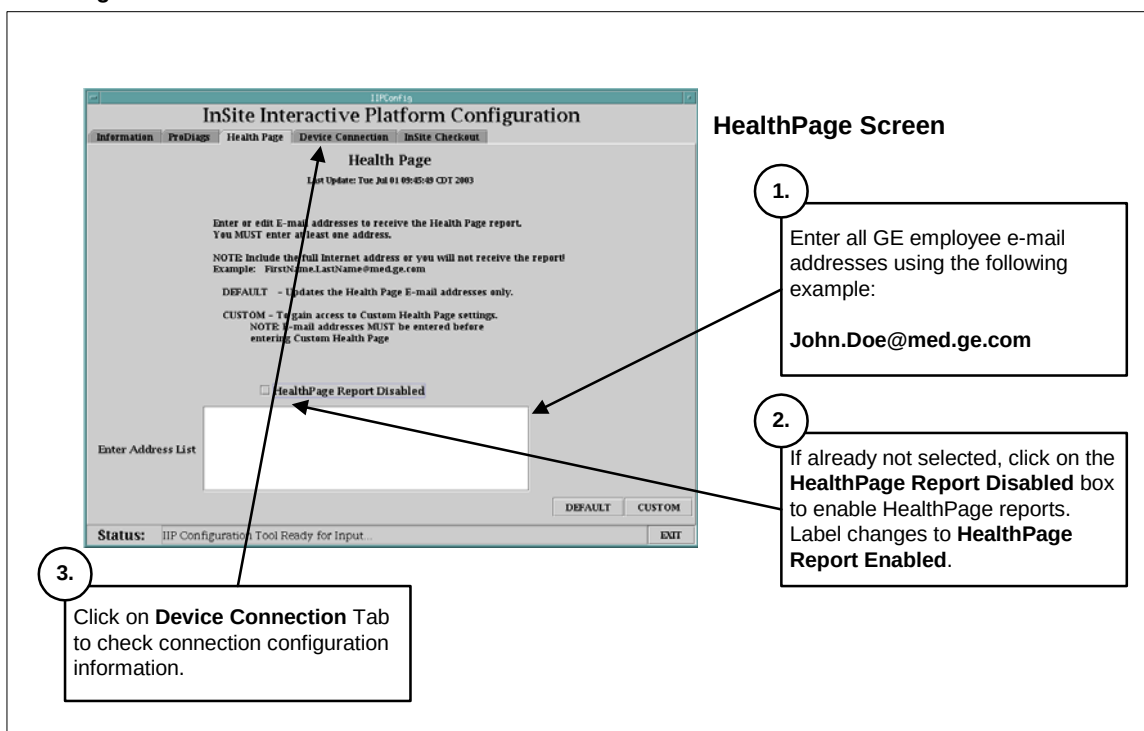
After completing the ProDiags setup, the IIP Admin Configuration utility automatically switches to the HealthPage Tab as shown in Figure 4.

The Restore State process (after loading Class M software) restored the following information for the HealthPage Reports:

- All e-mail addresses of targeted recipients that were previously loaded
- HealthPage report contents that was previously configured
- HealthPage report schedule that was previously configured

Unless you have recent changes to any of these three items, you can skip the HealthPage Tab of the IIP Admin Configuration utility and select the Device Connection Tab. If you have changes that you want to make (e-mail addresses, report content, report schedule), follow the instructions on the IIP Admin Configuration utility.

**Figure 4**  
**Insite Interactive Platform Configuration Utility**  
**HealthPage Tab**



## 6.5 Check Device Connection Setup

### 6.5.1 Modem Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen as shown in Figure 5.

If your system was configured for Modem connectivity, the Restore State process (after loading Class M software) restored all the Modem setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

- If there were no Modem parameters in the .insiteINFO file but there were BroadBand parameters, skip to the Check BroadBand Device Connection Setup procedure.
- If there were Modem parameters in the .insiteINFO file but the parameters displayed in the Modem Setup Screen do not match what you previously recorded, enter the correct values as required.

**NOTE:** The Modem configuration parameters should be pre-loaded from the Restore State process. Do not change settings unless these values were not properly restored.

If required, follow the instructions in Figure 5. The default values are usually the correct values or most commonly used. Parameter selections that do not require customization are:

CPU Serial Port Name

**Figure 5**  
**Insite Interactive Platform Configuration Utility**  
**Device Connection Tab – Modem Setup**

### Device Connection Modem Screen

**InSite Interactive Platform Configuration**  
Information | PreDiags | Health Page | Device Connection | InSite Checkout  
Last Update: Tue Jul 01 05:05:50 CDT 2003

Device Connection Type: Modem

Select the InSite Connection Type above, or select the modem settings given below and Press **APPLY** - To update the modem and modem configuration files with model and country defaults.  
**CUSTOM** - For other modem settings.

Dial-out Prefix: 9.1  
Modem Type: MultiTech MT5634ZBA V.90  
CPU Serial Port Name: /dev/ttyS0

Dialing mode: Tone  
Country: Default - All Others  
CPU Serial Port Speed: 38400

**APPLY** **CUSTOM** **EXIT**

Status: Reading Modem Configuration Files...

1.

Select the Dial-out Prefix value recorded from the previous .insiteINFO file (INS\_DialPrefix) from the pull down menu. This value is the number you must use to dial out of the hospital facility.

2.

Select the Modem Type value recorded from the previous .insiteINFO file (INS\_ModemType) from the pull down menu.

3.

Select the Dialing Mode value recorded from the previous .insiteINFO file (INS\_ToneOrPulse) from the pull down menu.

4.

Select the Country value recorded from the previous .insiteINFO file (INS\_Country) from the pull down menu.

5.

Select the CPU Serial Port Speed value recorded from the previous .insiteINFO file (INS\_TtySpeed) from the pull down menu.

We recommend that you select the highest Port Speed selection available (**115200**) regardless of the previous selection. Making this selection helps modem connected systems maximize connectivity performance.

6.

Select the CPU Serial Port Name value recorded from the previous .insiteINFO file (INS\_TtyPort) from the pull down menu.

7.

Click on **Apply** to set configuration parameters and display the Insite Checkout Tab.

### 6.5.2 BroadBand Setup

If your system was previously BroadBand Insite connected, click on the Network button of the IIP Admin Configuration utility Device Connection Tab to display the Network Setup Screen as shown in Figure 6.

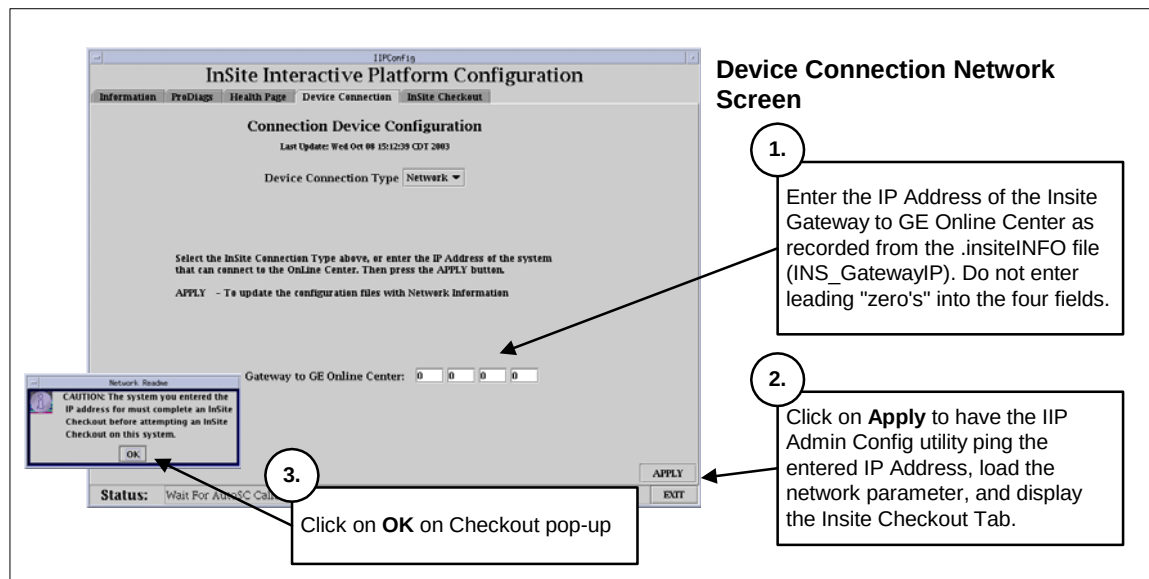
If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

**NOTE:** The BroadBand Network configuration parameters should be pre-loaded from the Restore State process. However, the on-site FE who performed the original Insite Checkout of the system may have used the hospital's imaging LAN default gateway IP Address (selection was available in the previous release of IIP).

**DO NOT USE THE HOSPITAL WORKING LAN DEFAULT GATEWAY IP ADDRESS AS THE INSITE GATEWAY IP ADDRESS OTHER THAN AS A TEMPORARY MEANS TO CONNECT TO THE ON LINE CENTER!!**

To find the correct Insite Gateway IP Address, use the contacts listed in Section 1.10, BroadBand Insite Connected Systems.

**Figure 6**  
**Insite Interactive Platform Configuration Utility**  
**Device Connection Tab – Network Setup**



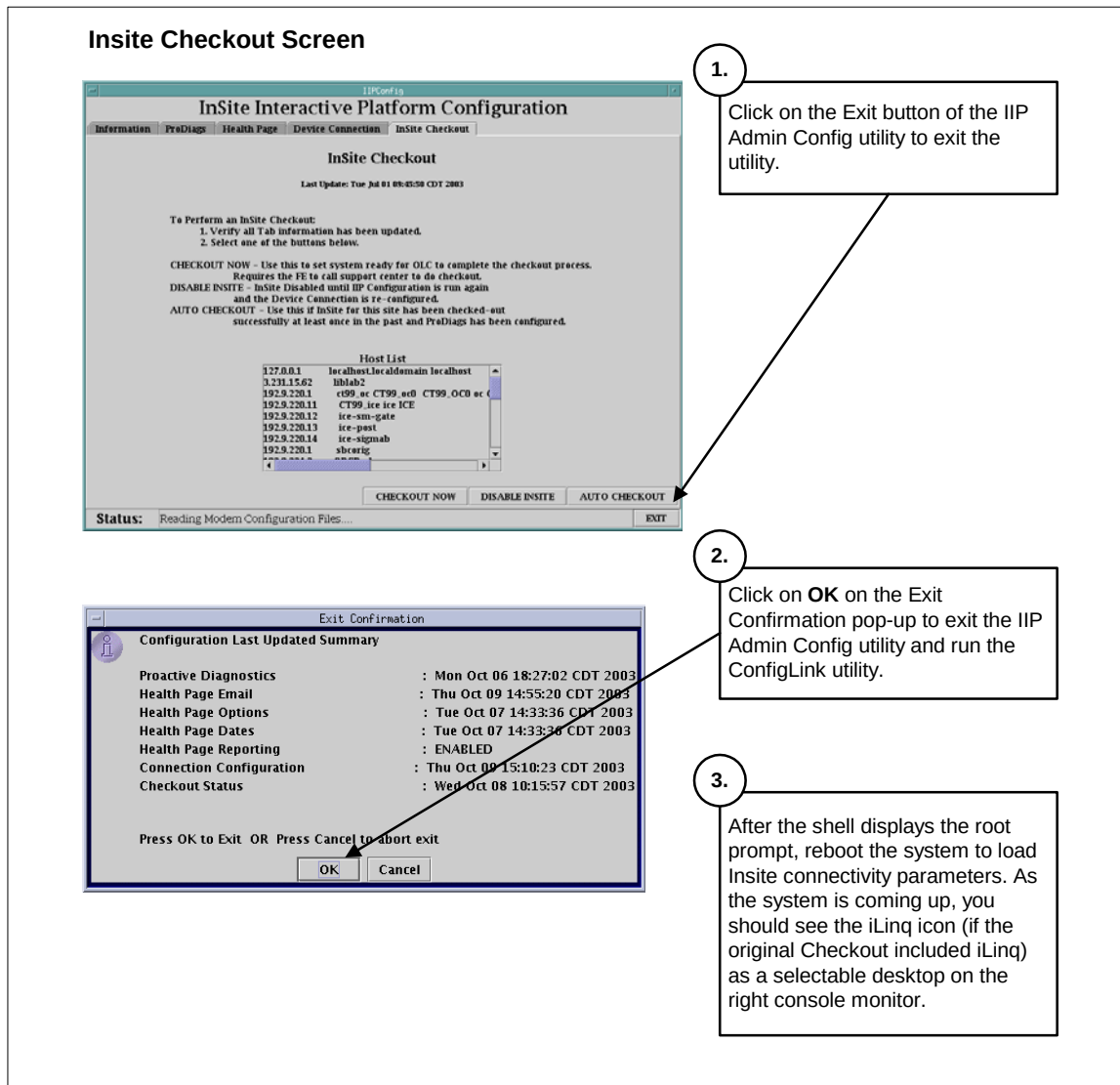
## 6.6 Completing the Insite Re-Initialization

Follow the instructions provided in Figure 7 to exit the IIP Admin Configuration Utility.

Figure 7

Insite Interactive Platform Configuration Utility

Insite Checkout Tab – Exiting the IIP Admin Configuration Utility



## 6.7 Run ConfigLink

Run the Insite **ConfigLink** script to populate all configuration information required to fully restore connectivity parameters on the system.

1. Open a shell.
2. Enter root as superuser:
  - a. At the ctuser prompt, type: `su -`
  - b. At the Password prompt, type: `#bigguy`
3. Change directory to Insite. At the root prompt, type:  
`cd /usr/g/insite`
4. Run the ConfigLink script. At the root prompt, type:  
`ConfigLink`

Wait for the shell to return the root prompt to ensure the **ConfigLink** process completes. When complete, exit root and close the shell.

## 6.8 Reboot System

To ensure all Insite configuration parameters are properly loaded into the operating system (route tables, etc.), perform a system reboot.

1. Click on the pink Shutdown button.
2. From the Shutdown selection pop-up, click on , then , to reboot the system.

## 6.9 Save System State

At the completion of the Load From Cold Process and Insite Re-Initialization process, ensure all system state parameters are saved using the Save System State Utility.

**Reference:** Direction 2378261-100, GRE Software Installation

Section 3.2      Save System State

## 8 Finishing Up

Perform the Gantry Rotation Safety check, found in the System Service Manual. If you cannot find the procedure, click on the link from the Tube Change procedure, as the manual may have a bug where it may not be displayed.

Also perform an application scan to ensure proper system operation before turning the system over to the customer.